MOD300 Anvendt Python programmering og modellering

Enrico Riccardi[1]

Department of Mathematics and Physics, University of Stavanger (UiS).[1]

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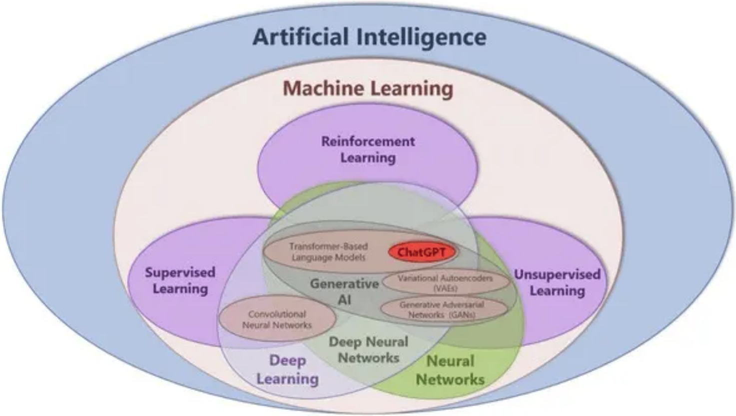
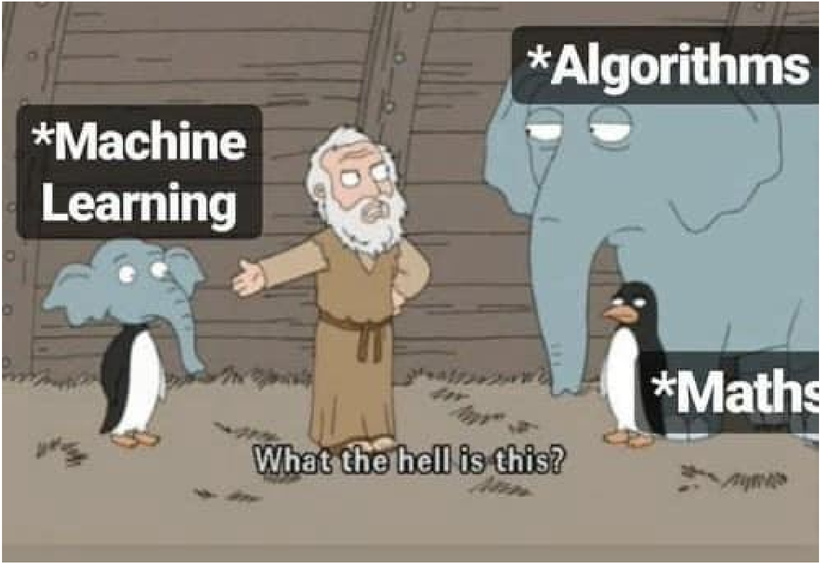
1. Statistics, Machine learning or Artificial intelligence?
2. Model evaluation
3. Reinforcement learning

Statistics, Machine learning or Artificial intelligence?

What is the main difference between the three fields?



How Machine Learning Started?



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- Statistics (origin "description of a state/country") is the discipline that concerns the collection, organization, analysis, interpretation, and presentation of data.
- It is conventional to begin with a statistical population or a statistical model to be studied. Populations can be diverse groups of people or objects such as "all people living in a country" or "every atom composing a crystal".
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Machine learning

Definitions:

- Machine learning is a branch of artificial intelligence (AI) and computer science which focuses on the use of data and algorithms to imitate the way that humans learn, gradually improving its accuracy. [IBM]
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One technical definition

Machine learning is a set of computer based statistical approaches that aim to minimise the loss function to maximise inference accuracy. [Enrico, 5.2.2024]

The loss function is the actual engine in machine learning.

Loss function

It quantifies the difference between the predicted outputs of a machine learning algorithm and the actual target values.

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Model evaluation

We can measure the accuracy of a prediction only when we have a 'truth' to compare with.

Model outcome

The accuracy of a prediction is limited by the available data set (not an universal quantity).

Given a dataset, we split the data in a train and a test set.

One has to be extremely careful to not introduce a **bias** in each of them when splitting. (or to introduce a bias properly, when for example, working with time series.

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Train

A train dataset is a subset of the original data. Generally it is circa a 70 % (common practice, not a rule) of the *INSTANCES*.

For each selected instance, all the features are kept, as their labels.

Random measurements

Some models are able to also consider the sequence of instances; the selection of the train dataset has to be randomised.

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Model evaluation

Train-Validation-Test

1. Data is split into train, validation and test dataset
2. A model is trained on the train dataset
3. The model (hyper)parameters are improved on the validation dataset
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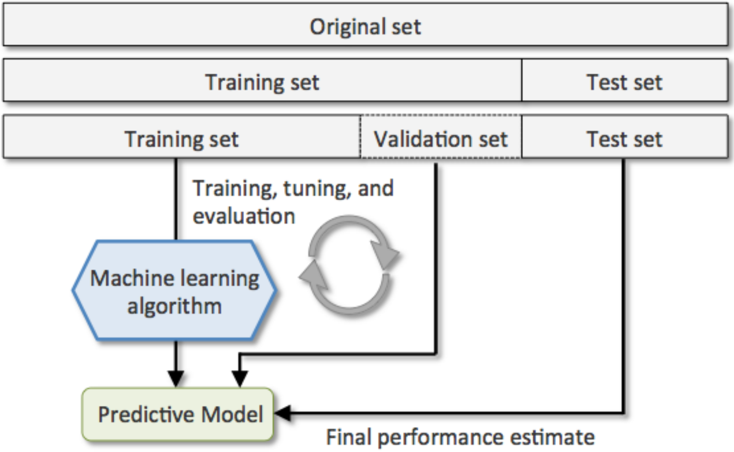

Cross validation

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1. Data is split into train and test.
2. Train dataset is then split into k-folds (different subsets, usually 5 to 10)
3. A model is trained on all k subset but one, sequentially
4. The model is tested on the remaining k data subset and its output averaged
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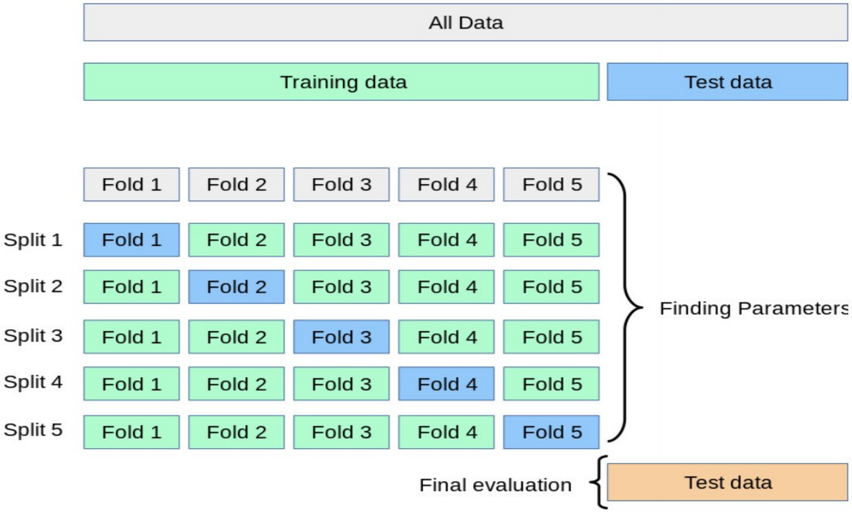
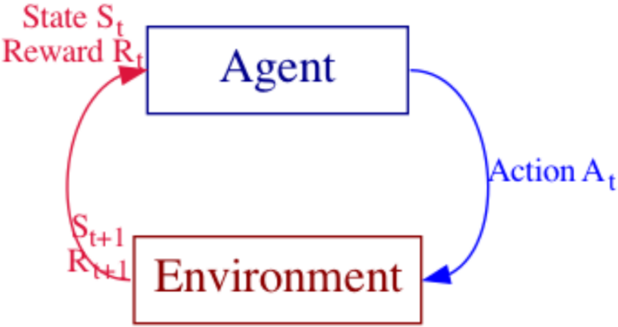


Reinforcement learning



Main definition

Loop of

State, action, reward

1. You evaluate the state where you are
2. you pick an action
3. you get a reward

Environment, State, Policy, Agent, Reward

Environment

Physical world in which the agent operates

State

Current situation of the agent

Agent

Decision Maker that take action according to a policy

Policy

Method to map agent's state to actions

$$\pi(a|s) = P(A_t = a|S_t = s)$$

Reward

Feedback from the environment

Episode

As Series of Atomic Experiences

Math backbone

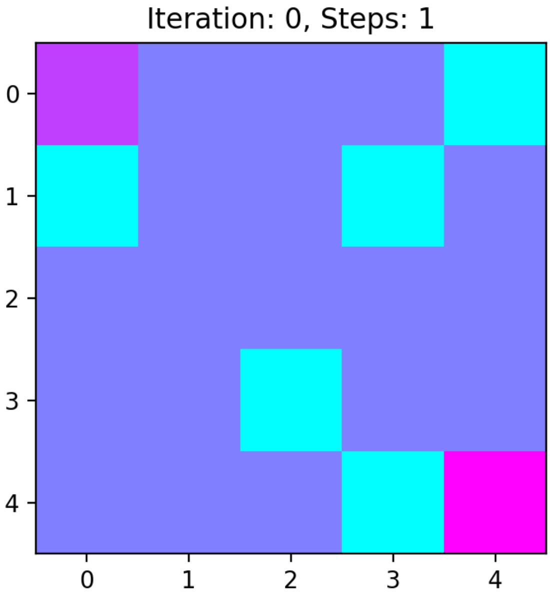
Q-learning

$$Q^{new}(S_t, A_t) = (1-\alpha)Q(S_t, A_t) + \alpha(R_{t+1} + \gamma max_{\alpha}(S_{t+1,\alpha})$$

where:

- α is the learning rate
- γ is the discount factor
- $max_{\alpha}(S_{t+1,\alpha})$ is the estimate of the optimal future value
- R is the reward
- A is the action
- S is the state

Example of RL



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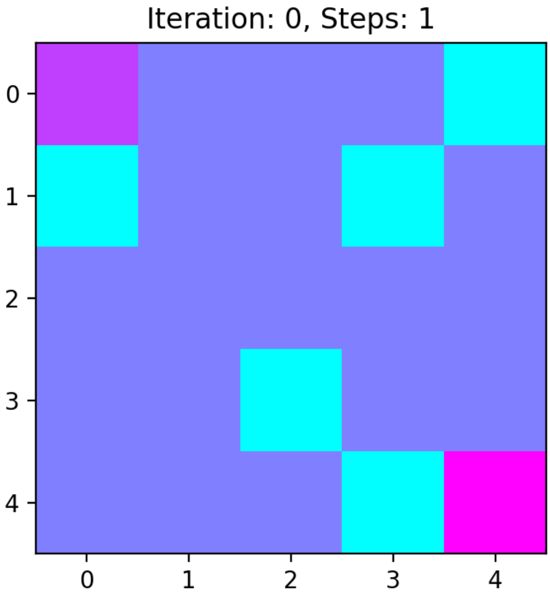
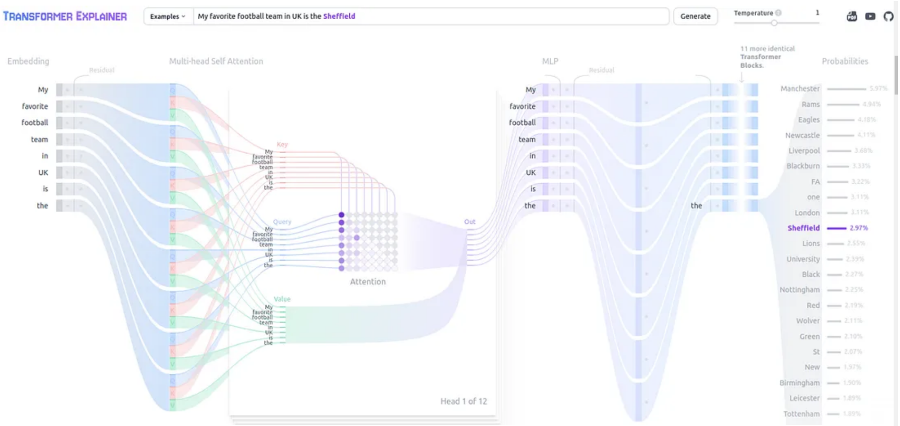


RL in LLMs

Pre-training

First goal: the next world prediction



How transformers work (Phase 1): A visual guide — Image Source: Author, adapted from the public website https://poloclub.github.io/transformer-explainer/"

Supervised Fine-tuning (SFT)

Language is inherently flexible - there are many valid ways to respond to any prompts

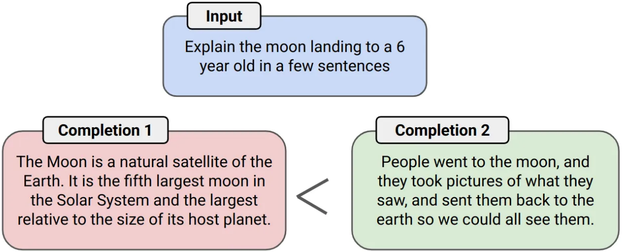
Many responses have been provided by humans. Multiple trainers in multiple countries.

Prompt	Response
Serendipity means the occurrence and development of events by chance in a happy or beneficial way. Use the word in a sentence.	Running into Margaret and being introduced to Tom was a fortunate stroke of serendipity.
ELI5: What's the cause of the "anxiety lump" in our chest during stressful or disheartening experiences?	The anxiety lump in your throat is caused by muscular tension keeping your glottis dilated to maximize airflow. The clenched chest or heartache feeling is caused by the vagus nerve which tells the organs to pump blood faster, stop digesting, and produce adrenaline and cortisol.
Create a shopping list from this recipe: Trim the ends off zucchini. Cut zucchini in half lengthwise; scoop out pulp, leaving 1/2-in. shells. Finely chop pulp. In a skillet, cook beef, zucchini pulp, onion, mushrooms and peppers over medium heat until meat is no longer pink; drain. Remove from the heat. Add 1/2 cup cheese, ketchup, salt and pepper; mix well. Spoon into the zucchini shells. Place in a greased 13x9-in. baking dish. Sprinkle with remaining cheese.	Zucchini, beef, onion, mushroom, peppers, cheese, ketchup, salt, pepper

RL in LLMs

Human feedback

Comparative feedback between answers.



Humans label which completion is preferred.

State, Evaluate, Reward loop in LLMs

The agent is the LLM: It makes a few prompt

Human say which one is the best

LLMs generates the text such that maximize the feedback